

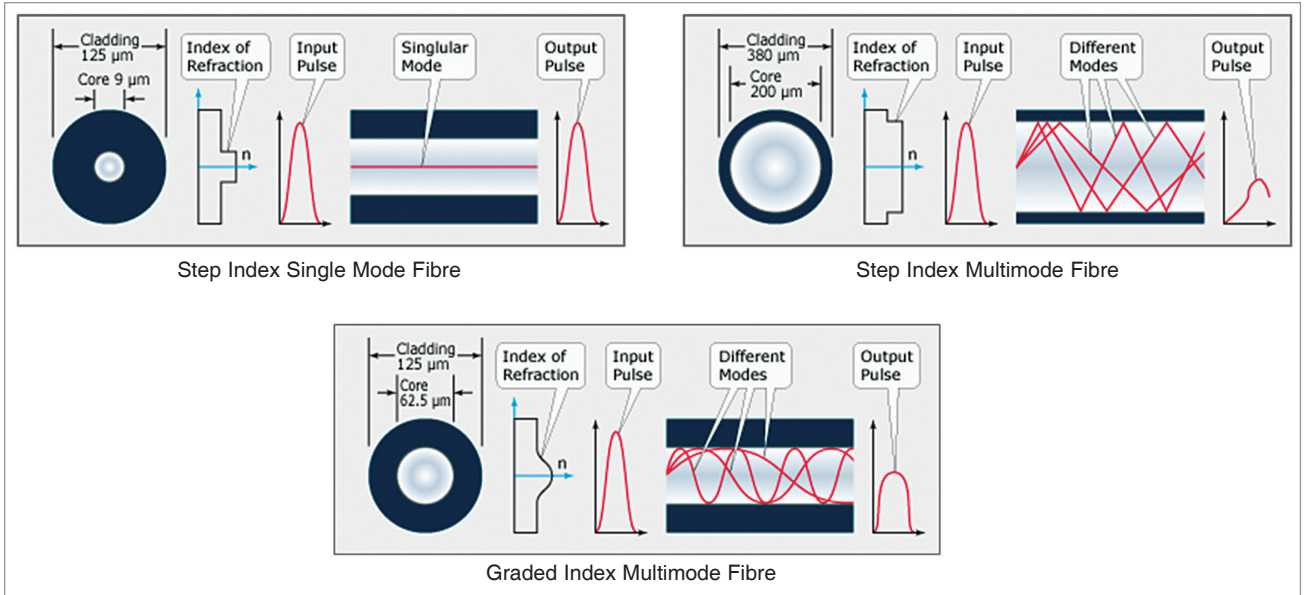


Fig. 2: Optical fibre classification (Source: <https://www.cablewholesale.com>)

traditional single-core optical fibre, enabling the transmission of multiple signals simultaneously. MCF's design increases fibre density and bandwidth while maintaining the compact size of the optical fibre.

Based on core pitch (core-to-core distance), MCF can be classified into three types:

Strongly coupled MCF. Cores are placed very close to each other, causing significant optical coupling between them

Weakly coupled MCF. Cores are spaced apart, resulting in lower crosstalk between cores

Randomly coupled MCF. Cores are placed randomly within the cladding, causing irregular coupling

characteristics. These MCFs are used in specialised applications

A 19-core MCF has been developed with a standard cladding diameter of 125µm. This fibre has been tested over a 63.5km distance, resulting in a data transmission speed of 1.7Pb/s. This MCF was developed through a collaborative project involving the National Institute of Information and Communication Technology (NICT), Japan, Sumitomo Electric Industries, Japan, and Macquarie University, Sydney.

An MCF designer must optimise core design, the number of cores, core arrangements, outer cladding thickness, and cladding diameter to balance the optical and mechanical

properties of the fibre. Due to the complexity of parameter optimisation and manufacturing, any MCF deployment system requires fast, low-loss, and high-strength splicing capabilities for effective deployment.

Advantages of MCF over SCF

Key advantages of MCF over single-core fibre (SCF) include:

Increased data transmission capacity. MCF can transmit multiple signals simultaneously through its multiple cores, providing higher data transmission capacity than SCF

Reduced fibre footprint. Replacing multiple SCFs with MCF can reduce the physical space required for cabling in optical communication

infrastructure

Cost efficiency of an optical communication system. Although MCF production costs are higher than SCF, MCF can significantly reduce the overall cost of optical communication systems

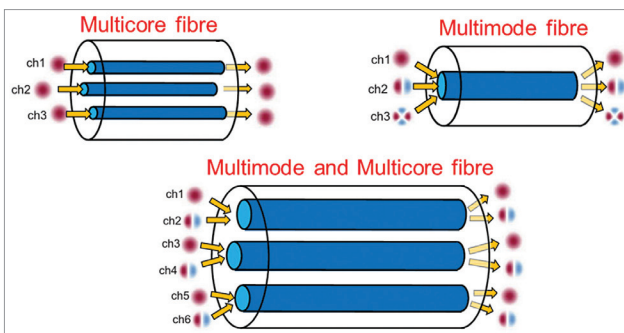


Fig. 3: Techniques of SDM (Source: <https://ieeexplore.ieee.org>)

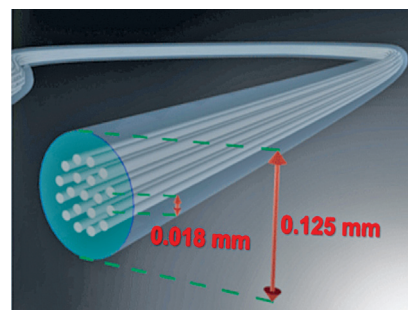


Fig. 4: Image of the developed 19-core optical fibre (Source: <https://sumitomoelectric.com>)